$c=\frac{1}{\sqrt{\mu_0\varepsilon_0}}$

No.

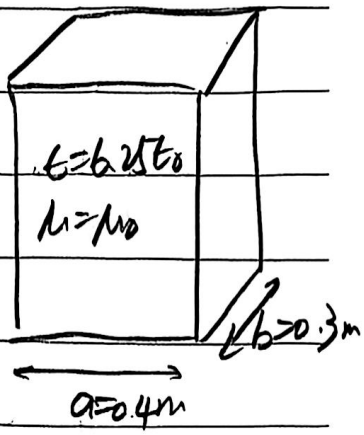
Date.

# 功能24 回忆卷

P1 (1) $k_x a=m\pi$, $k_x b=n\pi$

$k_c^2=k_x^2+k_y^2=\omega^2\mu\varepsilon$

$f=\frac{1}{2}\sqrt{\frac{1}{\mu\varepsilon}\left[\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]}=\frac{1}{2}c\sqrt{\left[\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]}$




$m=1, n=0$ 时, $f=0.15GHz$, $TE_{10}$

$m=0, n=1$ 时, $f=0.2GHz$, $TE_{01}$

$m=1, n=1$ 时, $f=0.25GHz$, $TE_{11}, TM_{11}$

(2) What is fundamental mode...?

$f>0.15GHz$?

P2 (1) $|F(\mu)|=\left|\frac{\sin\frac{N\mu}{2}}{\sin\frac{\mu}{2}}\right|=\left|\frac{\sin\frac{N\mu}{2}}{\sin\frac{\mu}{2}}\right|$

(2) $|F(\mu)|=\left|\frac{\sin\frac{N\mu}{2}}{\sin\frac{\mu}{2}}\right|$, $\mu=kd\cos\psi-\alpha$

$d=\frac{\lambda}{2}\Rightarrow kd=\frac{2\pi}{\lambda}\frac{\lambda}{2}=\pi\Rightarrow\mu=\pi\cos\psi-\frac{\pi}{2}$ 故 $\mu_{max}=\frac{\pi}{2}$

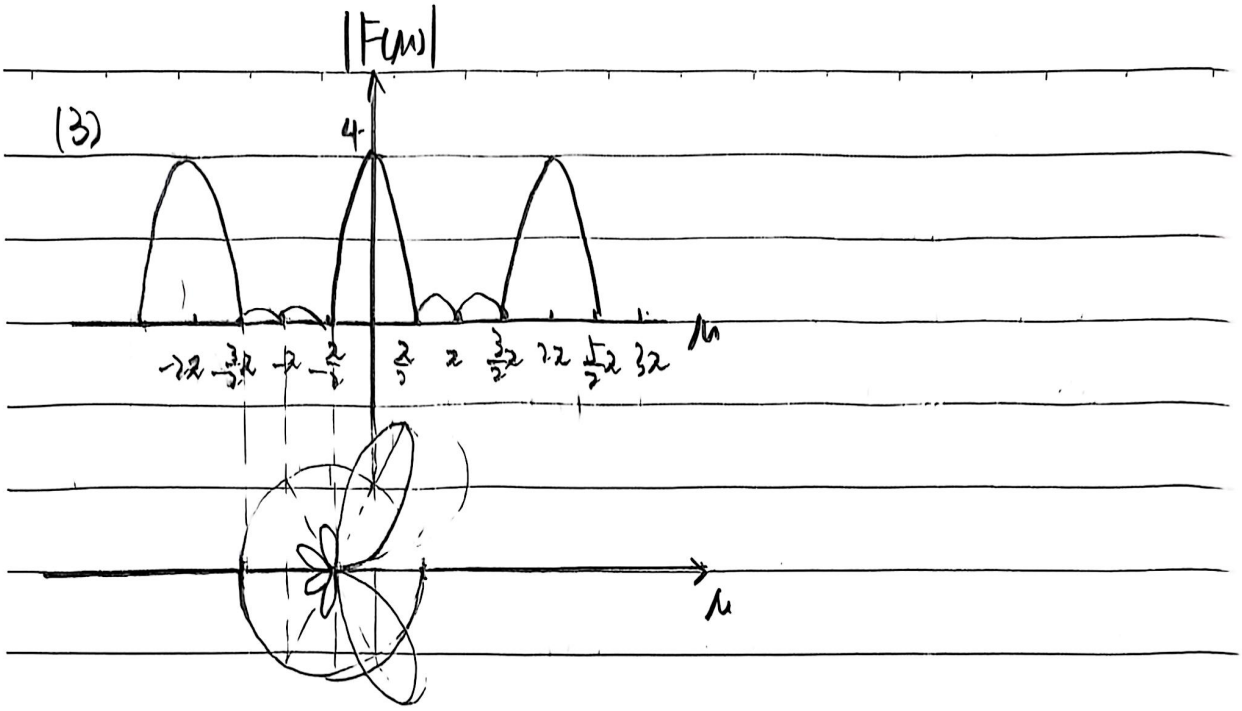
书(5.4.7) $\mu_{max}=2m\pi$? $\Rightarrow \mu_{max}=0$

$|F(\mu)|_{max}=N=4$

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(3)




(4) $\mu = x\cos\psi - \alpha$

$(\psi = \frac{\pi}{6} \Rightarrow \psi = \frac{\pi}{6})$

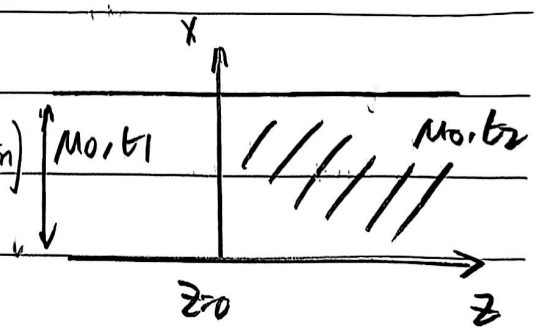
$\mu = \frac{\sqrt{3}}{2}\pi - \alpha = 2k\pi \Rightarrow \alpha = \frac{\sqrt{3}}{2}\pi - 2k\pi$

P3 (1) $\varepsilon_2 = \varepsilon_1 = 2\varepsilon_0$

$K_x d = m\pi$ $\left(d = \frac{\sqrt{3}\pi}{\omega\sqrt{\varepsilon_0\mu_0}} \Rightarrow d = \sqrt{3}\times 10^{-8}\,m\right)$

$K_0^2 = K_x^2 = \omega^2\mu_0\varepsilon_1$

$\Rightarrow f = \frac{\sqrt{2}mc}{2d}$




$d = \frac{\sqrt{3}\pi}{\omega\sqrt{\varepsilon_0\mu_0}}$, $f = \frac{\omega}{2\pi}$

$\varepsilon_1 = 2\varepsilon_0$, $\varepsilon_2 = \varepsilon_p = \varepsilon_0\left[1 - \frac{\omega_p^2}{\omega^2}\right]$

m=0时, $f = 0$GHz TM0

m=1时, $f = 12.24$GHz TM1, TE1

m=2时, $f = 24.49$GHz TM2, TE2

m=3时, $f = 36.74$GHz TM3, TE3

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(2) $\omega_p = \frac{1}{3}\omega \Rightarrow \Omega_2 = \frac{3}{4}60$

$\theta_c = \sin^{-1}\frac{k_4}{K} = \sin^{-1}\frac{\Omega_2}{\omega_1} = \sin^{-1}\frac{3}{8} = 22^\circ$

Operating Frequency ???

(3) $\alpha_b = \frac{1}{K_I}$

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